

MD. SHAHRIAR HOSSAIN

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 - **Portfolio:** <https://shahriar811.github.io/mdshahriarhossain.github.io/>
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SUMMARY

Computer Science and Engineering student specializing in Android development with experience building modern mobile applications using Kotlin. Skilled in designing intuitive user interfaces and developing scalable, high-performance apps with a focus on user experience. Familiar with integrating on-device AI and automation features, as well as working with machine learning models for real-world applications. Proficient in C, C++, Python, and Java, with a strong foundation in object-oriented programming and problem-solving. Eager to contribute to innovative mobile development projects and grow as a professional Android developer.

PROJECTS

Solvix AI – Offline On-Device AI Chatbot

1. Engineered a secure, offline AI chatbot for Android, enabling users to interact with large language models (LLMs) and execute multimodal tasks without requiring an active internet connection.
2. Implemented Google MediaPipe to deploy and run LLMs directly on-device, optimizing the application for edge computing, minimal battery consumption, and low-latency inference.
3. Integrated Google Firebase to build a robust and secure user authentication and login system.
4. Developed the application using Kotlin and Android Studio, ensuring 100% data privacy by architecting the system to process all user prompts and data entirely on the local device.
5. Technologies Used: **Kotlin, Android Studio, Google MediaPipe, Google Firebase, On-Device LLMs, Edge AI Computing.**

CineLog – Movie Discovery & Review App

1. Developed a modern Android application using Kotlin and Jetpack Compose, delivering a highly responsive and visually consistent user interface with Material 3 design principles.
2. Architected a scalable codebase utilizing the ViewModel and Repository patterns, integrating Hilt for robust dependency injection and Navigation Compose for seamless screen transitions.
3. Integrated the TMDB REST API via Retrofit, OkHttp, and Kotlinx Serialization to fetch and display trending movies and search results, using Coil for efficient, asynchronous image caching.
4. Implemented secure user authentication with Firebase Auth and Google Credential Manager, supporting both standard email/password and Google Sign-In flows.
5. Managed local user preferences, including a dynamic light/dark theme switcher, using DataStore Preferences for persistent storage.
6. Technologies Used: **Kotlin, Jetpack Compose, Firebase Auth, Retrofit, Hilt (DI), Material 3, REST APIs.**

Life Dashboard – Productivity & Finance App

- 1.Developed an offline-first, all-in-one productivity application using Kotlin and Jetpack Compose, featuring a centralized dashboard for task management, goal tracking, and daily expense monitoring.
- 2.Architected a robust local storage solution implementing Room Database for persisting complex data (tasks, goals, expenses) and DataStore for user preferences, adhering strictly to the MVVM design pattern.
- 3.Engineered precise background scheduling utilizing Android's AlarmManager and BroadcastReceiver to trigger exact, reliable notifications for user task reminders independent of the app's lifecycle.
- 4.Built responsive and reactive user interfaces leveraging Kotlin Coroutines and Flow for seamless asynchronous data streams, and integrated custom UI elements like interactive pie charts and live countdown timers.
- 5.Technologies Used: **Kotlin, Jetpack Compose, MVVM, Room Database, DataStore, Coroutines & Flow, AlarmManager, Navigation Compose.**

AuraPlay – Local Music Player Application

- 1.Developed a modern, lightweight music player using Kotlin and Jetpack Compose, delivering a clean and responsive Material 3 user interface with persistent theme preferences managed by DataStore.
- 2.Engineered seamless background audio playback by integrating the Media3 (ExoPlayer) library within a foreground MediaSessionService, ensuring uninterrupted listening and enabling external media controls.
- 3.Architected a robust local data layer utilizing Room Database to efficiently persist user-generated custom playlists, favorite songs, and locally scanned audio metadata from the device's MediaStore.
- 4.Optimized asynchronous operations and UI state management utilizing Kotlin Coroutines and Flow to handle real-time library searches, dynamic sorting, and efficient album art loading via Coil.
- 5.Technologies Used: **Kotlin, Jetpack Compose, Media3 (ExoPlayer), MVVM, Room Database, Coroutines & Flow, DataStore, Coil.**

SKILLS

Programming Languages: C, C++, Java, Python, Kotlin.

Tools & Core Concepts: Git/GitHub, Figma (UI/UX Design), Data Structures & Algorithms, Object-Oriented Programming (OOP), Machine Learning, Deep Learning, and LaTeX(Overleaf).

Project Management: Trello.

EDUCATION

- B.Sc. in CSE | North South University, Dhaka | Current (4th Year)
- HSC (Science) | Mymensingh College | 2020 (GPA: 5.00)
- SSC (Science) | Progressive Model School | 2018 (GPA: 4.89)

ADDITIONAL INFORMATION

- **Languages:** English, Bangla
- **Hobby:** Photography